Technical Report: Multi-Objective Shipment Route Optimizer

Balancing Freight Cost, Transit Time, and Disruption Risk Under Uncertainty

1. Introduction & The Practical Problem

When moving commercial freight across continents, selecting a shipping route is rarely as simple as picking the cheapest option on a rate sheet. In real-world logistics, the cheapest route almost always cuts corners: it runs through congested maritime bottlenecks, uses low-tier feeders, or transits geopolitical flashpoints like the Bab-el-Mandeb Strait or Suez Canal. While the upfront booking price looks attractive, an unexpected two-week delay can destroy cold-chain pharmaceuticals or shut down an automotive assembly line, resulting in late penalties and demurrage that cost five to ten times the initial freight savings.

The goal of this project was to build a decision-support engine that moves beyond simple heuristic sorting. I wanted a system that: (1) filters out mathematically inferior routes, (2) balances competing commercial objectives (Cost, Speed, Risk) based on explicit shipper profiles, (3) simulates realistic delay uncertainty before booking, and (4) explains its recommendations in plain, actionable language that a logistics operator on a dispatch desk can immediately trust.

2. How I Defined "Optimality" in this System

Different shipments have fundamentally different definitions of "optimal." A container of retail shoes can afford to wait a week to save $1,000, but an emergency shipment of aircraft replacement parts must move immediately regardless of fuel surcharges. Rather than relying on arbitrary weighted averages that mash disparate units (dollars, days, and risk percentages) together, I structured the decision pipeline into three clean stages:

> **Stage 1: Eliminating Inferior Options (Pareto Dominance)**
> Before ranking anything, the system checks for Pareto dominance across all candidate routes. Route A strictly dominates Route B if it is equal or better across Cost, Transit Time, Composite Risk, and Delay Probability, and strictly better in at least one metric. Any dominated route is discarded immediately. This guarantees that our candidate set sits entirely on the non-dominated frontier.

> **Stage 2: Multi-Criteria Trade-offs via TOPSIS**
> To rank the remaining Pareto-optimal routes according to operator priorities, I implemented *TOPSIS* (Technique for Order Preference by Similarity to Ideal Solution). TOPSIS projects all normalized criteria into vector space and identifies both a hypothetical "Dream Solution" (cheapest, fastest, safest) and a "Nightmare Solution" (costliest, slowest, riskiest). It scores routes by relative Euclidean closeness: `C_i = D_worst / (D_best + D_worst)`.

> **Stage 3: Quantifying the Hidden Cost of Delays (Monte Carlo Simulation)**
> Sticker price is a vanity metric in freight. To account for true operational risk, I built a 1,000-run Monte Carlo engine for every route. For each leg of a journey, delays are modeled as a Bernoulli trial (will a delay happen?) combined with a Log-Normal distribution for dwell duration. The engine computes the **Expected Realized Cost**—combining direct freight, late delivery contractual fines, and port demurrage—along with 95% Conditional Value-at-Risk ($CVaR_{95}$) to reveal severe downside exposure.

3. Synthetic Dataset Construction & Trade-Off Mechanics

Since live carrier booking APIs and private contract rates are proprietary, I constructed a synthetic dataset covering **40 international shipments** across 10 major global trade lanes (e.g., Shanghai–Rotterdam, Shenzhen–Los Angeles, Busan–Chicago, Mumbai–New York, Santos–Antwerp). I created realistic multi-leg journeys reflecting real-world shipping archetypes and their structural trade-offs:

- **Standard Ocean (Direct via Chokepoint):** Lowest direct freight cost ($3,200–$4,500), but carries heavy geopolitical exposure (0.84 risk) and a 42%–55% delay probability due to Red Sea/Bab-el-Mandeb disruptions, Suez convoys, and berth congestion.
- **Defensive Ocean (Cape of Good Hope Reroute):** Sails around southern Africa. Adds 8–10 days in transit and modest bunker fuel surcharges, but completely bypasses the Middle Eastern choke points, dropping geopolitical risk to 0.08 and cut delay chances to 16%.
- **Multimodal Sea-Air (via Dubai Hub):** Container vessel to Jebel Ali, followed by dedicated air cargo to European/US gateways. Delivers cargo in ~14 days at a balanced middle-tier price point.
- **Priority Direct Air Cargo:** Fastest delivery (2–3 days) with 95% on-time reliability, but priced at a 3x–5x premium with significant carbon intensity.
- **Smart Intermodal (Priority Berth + Dedicated Rail):** Combines guaranteed berth windows at coastal ports with electrified rail trunking, offering steady timelines and lower emissions.

4. Explainability: Giving Operators Context, Not Just Numbers

One of the quickest ways for dispatchers to reject an automated system is giving them a recommendation without explaining the rationale. I designed an Explainability Engine that translates multi-objective math into clear, operational English.

> **Example Output Generated for SHP-001 (Cold-Chain Pharma, Shanghai → Rotterdam):**
> *"Under the 'Balanced Trade-off' policy, 'Smart Intermodal with Guaranteed Berth & Priority Rail' is recommended for Pharmaceuticals & Cold Chain moving from Shanghai to Rotterdam. This route balances 21.5 transit days against a cost of $4,810.00 and a low risk score of 0.30. The cheapest alternative ('Standard Ocean via Primary Chokepoint' at $3,746.00) was rejected because it carries severe disruption risk (0.60) and a 54.6% delay probability, adding 4.0 days in transit. The fastest option ('Sea-Air via Dubai' at 14.0 days) was*

bypassed due to an excessive price premium of $1,958.00 (41% higher) which exceeds the required service level. The selected route is confirmed Pareto-optimal with an estimated 70.8% on-time reliability."

Additionally, the explainer computes concrete metrics: the **Marginal Cost per Day Saved** (e.g., $185 per day saved against the runner-up), flags vulnerable transit legs (e.g., Bab-el-Mandeb transit), and calculates a **Decision Stability Score** by testing whether the recommendation holds when fuel costs or priority weights shift by ±20%.

5. Empirical Evaluation: The "Cheapest-Only" Fallacy

To evaluate whether this multi-objective approach provides genuine value over traditional routing rules, I benchmarked our framework against three common industry baselines across all 40 shipments (160+ routes):

Routing Strategy	Pareto Rate	Avg Direct Freight	Avg Realized Cost	Avg Transit	Delay Prob	Composite Risk	OTIF Delivery	Regret
Proposed (Balanced)	**100.0%**	**$12,622**	**$16,068**	**12.0 d**	**20.7%**	**0.208**	**90.4%**	**0.740**
Proposed (Risk-Averse)	**100.0%**	**$24,462**	**$25,297**	**3.9 d**	**7.0%**	**0.152**	**97.5%**	**1.000**
Greedy Cost Minimizer	100.0%	$5,479	**$46,847**	19.7 d	53.4%	0.576	**49.7%**	1.242
Greedy Time Minimizer	100.0%	$24,462	**$25,297**	3.9 d	7.0%	0.152	**97.5%**	1.000
Random Baseline	100.0%	$11,247	**$49,621**	17.0 d	30.1%	0.318	**63.4%**	1.009

Observations and Discussion:

- **The Cheapest-Only Trap:** The Greedy Cost baseline looks best on paper ($5,478 direct freight). However, because it constantly routes through congested and high-risk waterways, its delay rate hits 53.4%. When contractual delay penalties and port demurrage kick in during simulations, its **average realized cost surges to $46,997**, and on-time arrival plummets to **49.7%**.
- **Balanced Policy Outperformance:** The proposed Balanced framework spends a reasonable upfront premium ($12,622 vs $5,478), but keeps expected realized cost down to **$16,069**—effectively saving over $30,000 per shipment in downstream disruption costs while delivering **90.4% on-time reliability**. It also achieves the lowest overall regret score (0.740).
- **100% Pareto Integrity:** In every simulated test, the optimizer selected a route on the true Pareto frontier, ensuring zero mathematically dominated options were ever suggested to the user.

6. Sensitivity & Robustness Testing

A practical routing system must not flip-flop its recommendation if fuel prices nudge up by 5% or transit times fluctuate slightly. I implemented a perturbation suite that stress-tests every recommendation across four operational shocks:

Stress-Test Scenarios Evaluated:

- **Scenario 1:** Fuel price spike (+20% cost weight)
- **Scenario 2:** Urgent delivery deadline (+20% time weight)
- **Scenario 3:** Geopolitical conflict flare-up (+20% risk weight)
- **Scenario 4:** Spot market freight discount (-20% cost weight)

Robustness Findings:
Across our 40-shipment benchmark, the Balanced recommendations maintained a **Decision Stability Index of 84.5%**. The decision boundaries proved resilient, only switching to expedited air or defensive Cape routes when operational priorities shifted dramatically.

7. Software Implementation & Architecture

I structured the project as a clean, modular Python codebase following production engineering practices:

- `src/models/schemas.py`: Strict Pydantic models for Legs, Routes, Shipments, and Risk matrices with computed properties.
- `src/optimization/`: Vectorized TOPSIS scoring, Pareto non-dominated sorting, and NumPy-based Monte Carlo simulations.
- `src/explainer/`: Algorithmic decision reasoning engine coupled with an optional Google Gemini copilot for conversational dispatcher briefings.
- `src/api/app.py`: Production-ready FastAPI service exposing `/recommend`, `/shipments`, and `/benchmark` endpoints.
- `app.py`: Interactive Streamlit web application featuring Pareto scatter charts, risk radars, and scenario playgrounds.
- `cli.py`: Rich terminal CLI for dispatchers to run recommendations and batch benchmarks from the command line.
- `tests/`: Automated test suite (13 tests) verifying Pareto properties, constraint filtering, and simulation invariants with 100% pass rate.

8. Limitations & Production Next Steps

> If I were deploying this into a live production environment at an enterprise scale, I would prioritize the following enhancements:
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> 1. **Live Telemetry Integration:** Static delay probabilities should be replaced with live AIS satellite feeds (e.g., MarineTraffic or Spire) and real-time port berth queue APIs.
> 2. **Dynamic Spot Rate Indexing:** Fuel prices and ocean container spot rates fluctuate weekly. Connecting to Platts Bunkerwire or Shanghai Containerized Freight Index (SCFI) feeds would improve cost accuracy.
> 3. **Carbon ETS Compliance:** Under the EU Emissions Trading System (EU ETS) for maritime transport, emissions have direct cash implications. Incorporating dynamic carbon pricing into the optimization objective vector would align routing with ESG mandates.

9. Conclusion

Evaluating global freight routes requires balancing multiple competing operational pressures under real uncertainty. By combining Pareto non-dominated filtering, TOPSIS multi-criteria ranking, and stochastic Monte Carlo delay modeling, this project provides an actionable, defensible routing system that prevents expensive disruption traps while providing logistics operators with clear, transparent reasoning.